Analysis of Dynamic Stall Mechanisms in Helicopter Rotor Environment



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A helicopter reaches its flight envelope limit at high-thrust forward flight or maneuver with high load factors because of dynamic stall. This phenomenon is due to complex unsteady three-dimensional flow separation mechanisms that occur on the retreating blade and can lead to structural damage of the pitch links. Because of the complexity of the flow conditions, a detailed description of dynamic stall in helicopter environment is still lacking. This paper proposes to investigate numerically an isolated 7A wind tunnel scaled rotor in a high-thrust forward flight. The objective is to make use of this numerical simulation to identify the mechanisms that lead to stall in this rotor configuration. Despite the high level of complexity of this flow, the comparison with experiment in terms of section normal forces and pitching moments is in very satisfactory agreement. The numerical results indicate three different separated flow regions, each one of them being characterized by different mechanisms. Furthermore, a blade–vortex interaction process is observed as a triggering factor of dynamic stall.

Nomenclature

a	freestream speed of sound, m/s
b	number of blades
C_L/σ	rotor lift coefficient, $F_Z/[\rho_\infty(R\Omega)^2 S\sigma]$
C_X/σ	rotor propulsive force coefficient, $F_X/[\rho_\infty(R\Omega)^2 S\sigma]$
C_f	skin friction coefficient, $\tau_w/[0.5\rho_\infty(r\Omega + U_\infty \sin\psi)^2]$
$C_m M^2$	section pitching moment expressed in the local airfoil frame
$C_n M^2$	section normal force expressed in the local airfoil frame
C_p	wall pressure coefficient, $(p - p_\infty)/[0.5\rho_\infty(r\Omega + U_\infty \sin\psi)^2]$
c	blade chord, m
F_X	rotor propulsive force in the wind frame, N
F_Z	rotor lift force in the wind frame, N
H_i	boundary-layer shape factor, ratio of displacement and momentum thicknesses
p	wall pressure, Pa
Q	Q-criterion, $0.5\,(\lvert\overline{\overline{\Omega}}\rvert^2 - \lvert\overline{\overline{S}}\rvert^2)$, where $\overline{\overline{\Omega}}$ and $\overline{\overline{S}}$ represent, respectively, the vorticity and the strain rate tensors
R	rotor radius, m
r	radial coordinate, m
S	rotor disk area, πR^2
U_∞	freestream velocity, m/s
α_q	shaft angle, deg
β_{1C}	longitudinal flapping angle, deg
β_{1S}	lateral flapping angle, deg
Δx	wall cell length in the chord direction, m
Δy	wall cell length in the wall-normal direction, m
Δz	wall cell length in the radial direction, m
θ_t	torsion angle at blade tip, deg
θ_0	collective pitch angle, deg
θ_{1C}	cyclic lateral pitch angle, deg
θ_{1S}	cyclic longitudinal pitch angle, deg
μ	advance ratio, $U_\infty/(R\Omega)$
μ_t/μ	turbulent-to-laminar viscosity ratio
ρ_∞	freestream fluid density, kg·m^{-3}
σ	rotor solidity, bRc/S
τ_w	wall shear stress, N·m^{-2}
ψ	blade azimuth, deg
Ω	rotor angular velocity, rad/s

Superscript

$+$	wall cell length (Δx, Δy, Δz) expressed in wall units

Introduction

Dynamic stall is one of the most difficult aerodynamic problems encountered on helicopter rotors. It is also a very challenging flow phenomenon for the computational fluids dynamics (CFD) community since it involves three-dimensional, unsteady, separated turbulent flows and fluid–structure coupling. Dynamic stall can occur on the main rotor blades during maneuvers and high-thrust forward flights. For these kinds of flight conditions, the equilibrium of the rotor requires to decrease the aerodynamic angle of attack of the advancing blade and to increase the

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angle of attack of the retreating blade. Thus, very high angles of attack can be reached on a large portion of the retreating blade side of the rotor disk, leading the boundary layer to massively separate from the blade suction side. This flow separation induces an increase of the drag, and as a consequence, of the fuel consumption. But most importantly, dynamic stall also generates large pitching moment oscillations of the blade that are responsible for large pitch link load variations that can lead to structural damage and the loss of pilot control. The flight envelope of helicopters is thus limited because of this dynamic stall phenomenon.

The complexity of dynamic stall encountered on helicopter rotor configurations led many researchers to first consider simplified configurations such as two-dimensional airfoils at low-frequency pitching motions (Ref. 1). Piziali provided an extensive data base of a NACA0015 airfoil in dynamic stall conditions (Ref. 2) that has been widely used to assess the predictive capabilities of different turbulence models of CFD approaches (Refs. 3–6). More recently, several wind tunnel test campaigns were devoted, at ONERA and DLR, to an OA209 airfoil, providing particle image velocimetry and unsteady wall pressure measurements (Refs. 7 and 8). These experimental data were used to validate unsteady Reynolds-averaged Navier–Stokes (URANS) simulation tools. This two-dimensional OA209 stall investigation allowed some requirements to be defined in terms of turbulence modeling and grid resolutions (Ref. 9). The influence of laminar–turbulent transition was numerically investigated in Refs. 10 and 11. An extensive analysis of the time resolution was proposed by Liggett and Smith for the case of VR7 airfoil in dynamic stall condition (Ref. 12). More recently, the configuration of a finite-span three-dimensional wing was numerically investigated by means of both URANS (Refs. 13–15) and zonal detached-eddy simulation (Ref. 16) approaches and validated with the experimental data of Ref. 8. The wing was composed of a single OA209 airfoil, with zero twist and an aspect ratio of 2.6. The numerical results showed a fairly good agreement with experiments and provided helpful insights into the dynamics of the flow separation around this oscillating three-dimensional wing. In Ref. 13, Costes et al. identified that stall first occurred in the inboard part of the blade. Then, a strong spanwise velocity appeared in the confined separated region close to the leading edge (LE). The separation region finally spread in the spanwise direction and contaminated a large part of the blade, whereas the boundary layer, in the very close region of the blade tip, was kept attached during the whole dynamic stall cycle because of tip vortex induced flow.

Based on these previous studies, this paper proposes to go further into the complexity of the flow configuration and to investigate a helicopter rotor flow in dynamic stall condition. The first experimental analysis of the characteristics of the dynamic stall on helicopter environment was done by Bousman, based on UH-60A flight tests (Ref. 17). Bousman specially identified multiple dynamic stall events on different radial and azimuthal stations of the rotor disk. Numerical simulations of dynamic stall on rotor configurations were investigated, first in 2006, by Potsdam et al. for a UH-60A flight-test configuration (Ref. 18), and more recently by Ortun et al. (Ref. 19) and Yeo et al. (Ref. 20) for the wind tunnel scaled 7A rotor. In these two last references, the simulations were performed by means of a loose coupling method between a CFD code and a comprehensive analysis (CA) code. U.S. Army used Helios (Ref. 21) for CFD and RCAS (Ref. 22) for CA, whereas ONERA used *elsA* (Ref. 23) for CFD and HOST (Ref. 24) for CA. Both approaches have showed their capabilities to accurately predict and understand structural loads of the 7A rotor in two flight conditions including dynamic stall. However, a detailed analysis of the flow separation mechanisms during dynamic stall is still missing. The objective of the present paper is to bring some new elements of understanding concerning the stall mechanisms on rotor environment by means of suitable postprocessing of numerical simulation.

After a brief introduction of the flow configuration and the numerical methods, the results of the numerical simulation are validated in the second section by comparison with the available experimental data. Then, a thorough analysis of the numerical results is proposed in the last section, to identify and characterize the mechanisms of flow separation and propose a scenario of dynamic stall in this rotor environment.

Flow Configuration and Numerical Methods

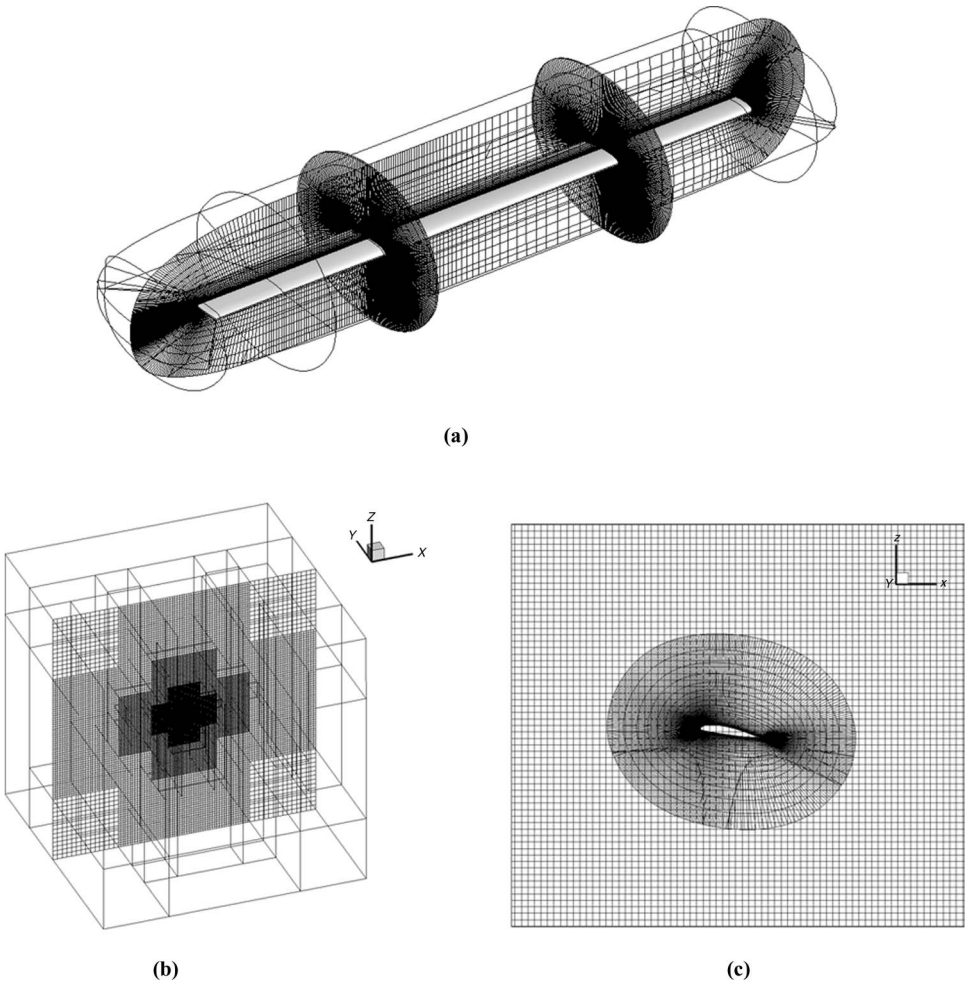
The flow configuration is taken from the 7A rotor data base, tested in the S1MA wind tunnel of ONERA, in Modane (France) in 1991 (Refs. 25 and 26). This rotor of radius $R = 2.1$ m is composed of four articulated blades of constant chord $c = 0.14$ m. For the inboard part of the blade ($r/R \leq 0.75$) a 13% relative thickness OA213 airfoil is used, whereas the 9% relative thickness OA209 airfoil is used in the tip region for $r/R \geq 0.9$. The aerodynamic twist of the blade follows a linear law with a constant rate of -8.3 deg/R. The ONERA S1MA is a transonic closed circuit atmospheric wind tunnel. The test section has a diameter of 8 m. The experimental database provides unsteady pressure measurements, integrated section normal forces, and pitching moments at the five blade radial stations $r/R = 0.5$, 0.7, 0.825, 0.915, and 0.975.

Dynamic stall has been experimentally observed on the 7A rotor for a forward flight condition at a moderate advance ratio $\mu = 0.3$ and a high lift coefficient $C_L/\sigma = 0.1$. This configuration is retained for the numerical investigation presented hereafter. The propulsive force coefficient is $C_X/\sigma = 0.0045$, and the rotor is trimmed to satisfy the so-called Modane flapping law: $\beta_{1S} = 0$ and $\beta_{1C} = -\theta_{1S}$. All the flow parameters are summed up in Table 1.

The computations are performed with the *elsA* solver, which discretizes the URANS equations on structured multiblock grids with a finite-volume approach (Ref. 23). A Chimera technique is adopted, with near body curvilinear grids around each blade and a multiresolution background Cartesian grid. The body meshes have an O-grid topology in the chord and spanwise directions and extend over one chord from the blade wall (Fig. 1(a)). The Cartesian grid is made of several near matching blocks with different levels of resolution (Fig. 1(b)). The Cartesian blocks closest to the body grids have a grid size length of 10% of chord (Fig. 1(c)). The test-stand is not taken into account in this simulation. The computational domain extends from $-25R$ to $25R$ in the three directions around the rotor. Since Ortun et al. showed that the wind tunnel walls have no effect on the blade airloads (Ref. 19), these are not modeled in the simulation and far-field boundary conditions are imposed in the external edges of the computational domain. The spatial resolution characteristics of the simulation are shown in Table 2. Each blade mesh is composed of 5.7 million points. The total number of points including the body grids of the four blades and the background Cartesian grid is 43.6 million points. The near-wall grid resolution on the blade has been made to be as close as possible as the requirements proposed in Ref. 13 for the configuration of an oscillating wing. In the chordwise direction, $\Delta x/c$ reaches 0.11% at the LE and 1.6% at midchord, where Costes et al. recommend 0.05% and 0.45% in Ref. 13. The spanwise resolution

Table 1. Flow conditions

Air density, ρ (kg/m^3)	1.018
Temperature, T (°C)	27.9
Rotor speed, Ω (rpm)	1022
Tip Mach number, $M_{tip} = R\Omega/a$	0.646
Advance ratio, μ	0.3
Rotor lift coefficient C_L/σ	0.1
Rotor propulsive force coefficient C_X/σ	0.0045



(a) (b) (c)

Fig. 1. View of mesh: blade grid (a), whole Chimera background grid (b), and overlapping region around a blade section (c).

Table 2. Comparison of the spatial resolution between the 7A rotor mesh and the finite-span wing mesh of Ref. 13

	7A Rotor	Finite-Span Wing (Ref. 13)
Number of points per blade	5.7 M	
Total number of points	43.6 M	
N around airfoil	313	977
N over the blade span	155	
$\Delta x/c$ at leading edge	0.11%	0.05%
$\Delta x/c$ at midchord	1.6%	0.45%
$\Delta z/c$	5.6%	2.3%
$\max(\Delta y^+)$	0.4	0.6
Number of points inside the boundary layer at midchord and $r/R = 0.8$	40	35–50
Δx^+ at leading edge	130	$\leq$300
Δx^+ at midchord	800	$\leq$300
Δz^+ ar leading edge	7000	$\leq$3000
Δz^+ at midchord	3000	$\leq$3000

is refined in the blade tip region to reach $\Delta z = 5.6\%$ at $r/R = 0.8$. More details about the influence of the grid resolution can be found in Ref. 32.

The time derivative is discretized with a second-order implicit gear scheme. At each time step, the nonlinear problem is solved by an iterative Newton process. The azimuthal time step is set to $\Delta\psi = 0.3$ deg, which corresponds to 1200 time steps per cycle. The number of subiterations of the Newton process is fixed to $N_{\text{sub}} = 30$ as suggested by the numerical parameters sensibility study of Ref. 32.

Following the recommendations of the previous two-dimensional (Refs. 4, 9, and 10) and three-dimensional (Refs. 13–15) dynamic stall numerical studies, the Kok version of the k–ω turbulence model (Ref. 27) coupled with the shear-stress transport (SST) correction proposed by Menter (Ref. 28) is chosen because of its satisfactory capability to capture boundary layer separation. No transition modeling is involved in the current study although this could have an influence of dynamic stall, at least for the innermost part of the blade where the local Mach and Reynolds numbers are expected to be low (Refs. 10 and 11).

The blade kinematics and deformations are taken into account by means of a loose coupling strategy between *elsA* CFD solver (Ref. 23) and HOST for CA (Ref. 24). At each rotor revolution, the loads provided by the CFD are used by HOST to compute the trim control angles and the deformations of the blades, which are in return used by *elsA* to impose the motions and the deformations of the blade grids. The simulations are stopped once the rotor control angles reach an equilibrium state. More details about this coupling strategy can be found in Refs. 29 and 30.

Validation

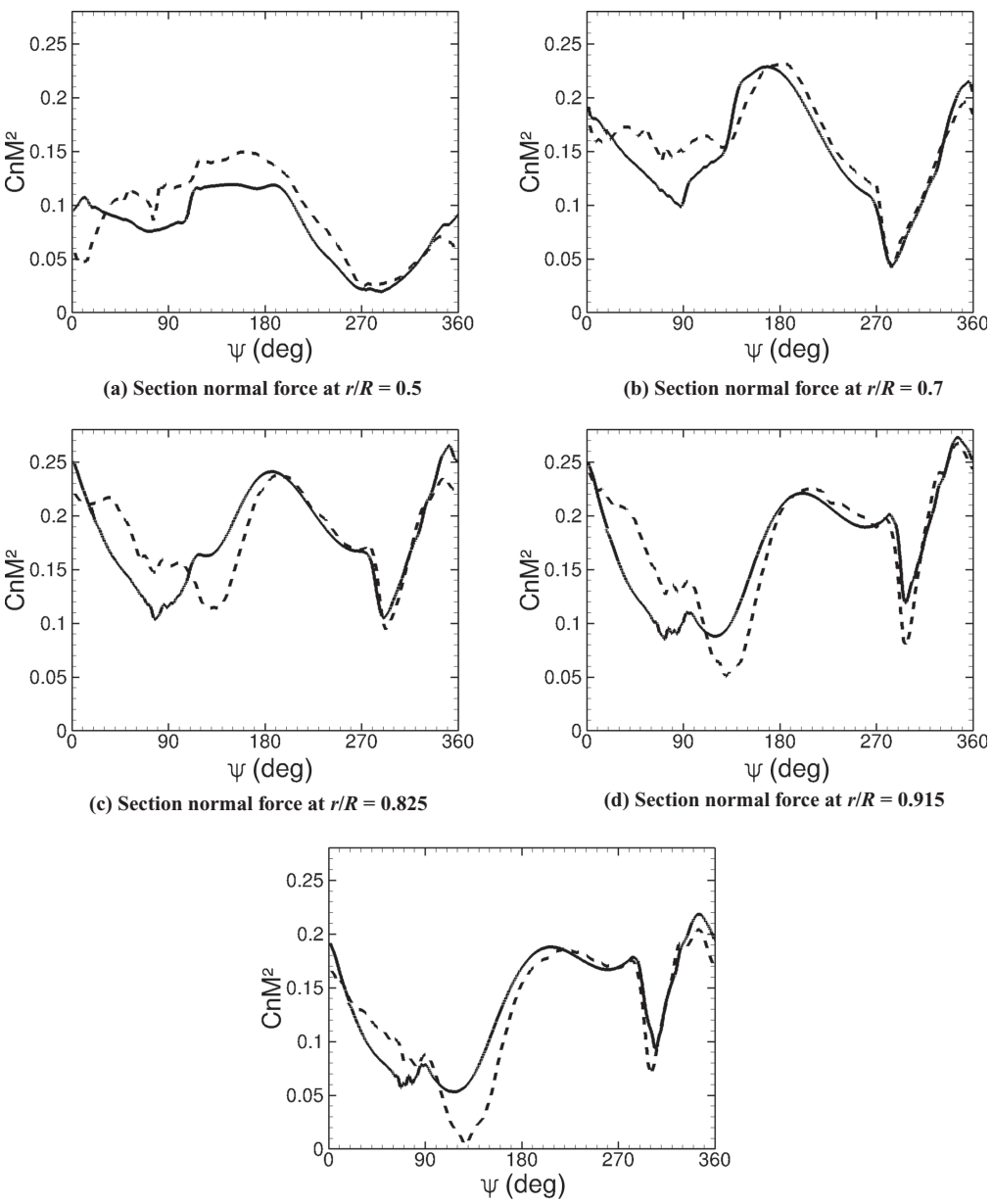
The final trim control angles obtained at the convergence of the *elsA*/HOST coupling simulation are compared to the experiment and

Table 3. Comparison of the trim control angles between experiment, HOST stand-alone simulation and *elsA*/HOST coupling simulation

	α_q	θ_0	θ_{1C}	θ_{1S}
Experiment	−6.7	8.4	3.16	−3.51
HOST	−6.1	7.66	1.51	−3.05
elsA/HOST	−6.40	8.41	2.26	−3.50

HOST stand-alone results in Table 3. The CFD/CA coupling method clearly improves the prediction of the control angles, compared to the pure CA approach. HOST underestimates the shaft angle and the collective pitch angle by 0.6 and 0.8 deg, respectively, where the CFD provides a good agreement with experiment. The longitudinal pitch angle θ_{1S} obtained with *elsA*/HOST matches the experimental value, whereas it is overestimated by 0.5 deg with HOST. A small disagreement between CFD and experiment is only observed for the lateral pitch angle, θ_{1C}, which is slightly underestimated in the simulation. As shown in Refs. 19 and 32, the disagreement on the lateral pitch angle can be overcome if the test-stand is included in the simulation. Indeed, the test-stand deflects the flow in the rotor disk. This creates upwash on the front of the rotor ($\psi = 180$ deg) and downwash on the rear part ($\psi = 0$ deg). Thus, it is necessary to increase the lateral pitch angle θ_{1C} to balance the decrease of the effective angle of attack induced by the downwash at $\psi = 0$ deg and the increase of the effective angle of attack induced by the upwash at $\psi = 180$ deg.

The section normal forces and pitching moments predicted by the simulation are compared to the experimental measurements in Figs. 2 and 3, respectively. The evolution of C_nM^2 is in fairly good agreement with experiments on the region of interest, i.e. the retreating blade side where stall occurs. The drop of C_nM^2 due to dynamic stall is quite well-captured by the simulation. The agreement between simulation and experiment on



(a) Section normal force at $r/R = 0.5$

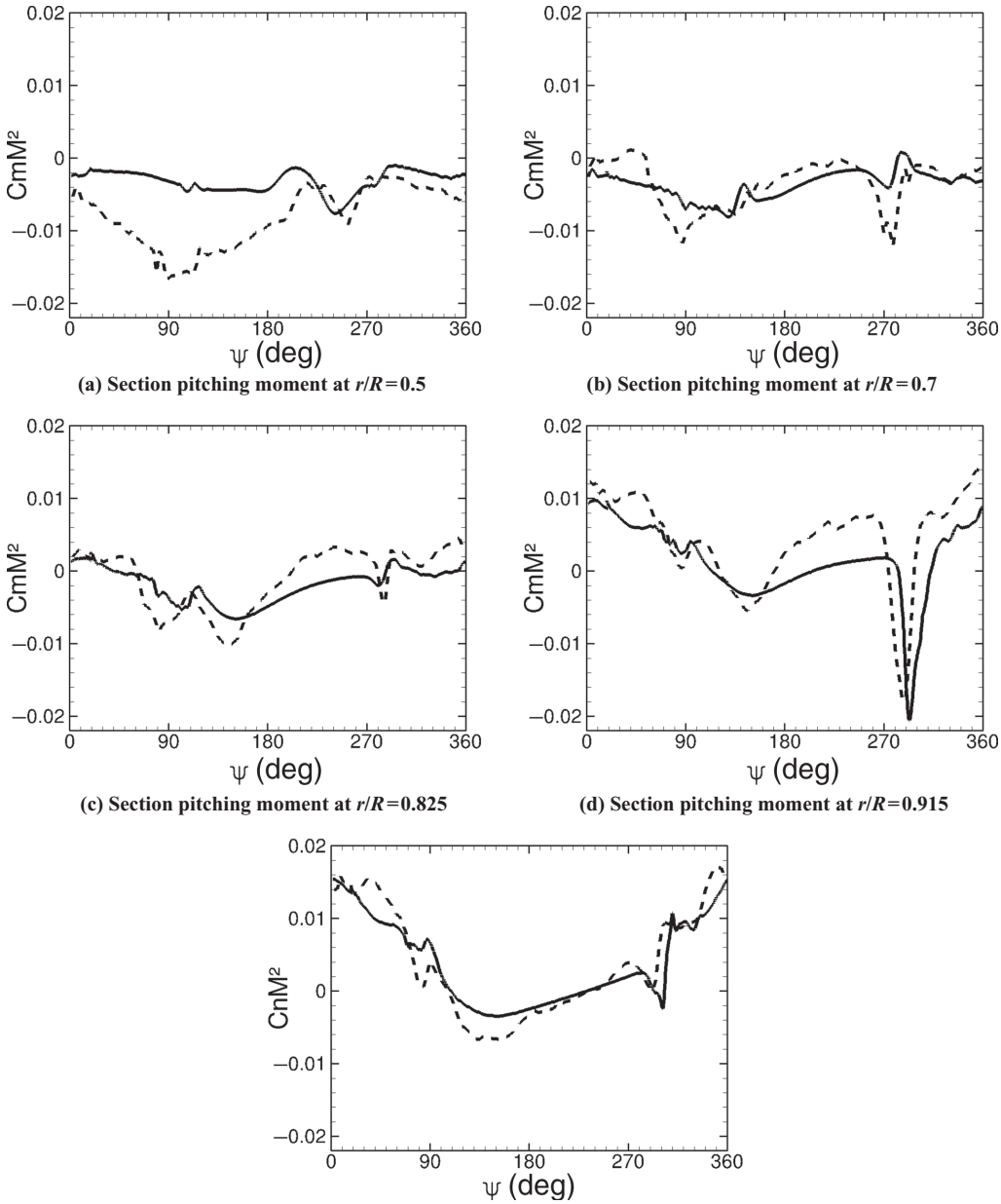
(b) Section normal force at $r/R = 0.7$

(c) Section normal force at $r/R = 0.825$

(d) Section normal force at $r/R = 0.915$

(e) Section normal force at $r/R = 0.975$

Fig. 2. Section normal force C_nM^2: Experiments (dashed line) and *elsA*/HOST coupling simulation (solid line).



(a) Section pitching moment at $r/R=0.5$

(b) Section pitching moment at $r/R=0.7$

(c) Section pitching moment at $r/R=0.825$

(d) Section pitching moment at $r/R=0.915$

(e) Section pitching moment at $r/R=0.975$

Fig. 3. Section pitching moment C_mM^2 : Experiments (dashed line), *elsA*/HOST coupling simulation (solid line).

the retreating blade is almost perfect at $r/R = 0.7$ and 0.825. For sections $r/R = 0.915$ and 0.975, although the simulation slightly overestimates the minimum of C_nM^2 reached at stall, the agreement with experiment can still be considered as fairly satisfactory. On the pitching moments (Fig. 3), CFD results provide oscillations of C_nM^2 at stall, which are more or less in good agreement with experiments depending on the sections. At midspan ($r/R = 0.5$), the phase when moment stall occurs is well captured by the simulation. At sections $r/R = 0.7$ and 0.825, CFD results give poor agreement of C_mM^2 with experiment, although some variations appear around the azimuthal range where moment stall is experimentally observed. At section $r/R = 0.915$, the pitching moment drop is the strongest in both CFD and experiment. The numerical simulation predicts a slightly delayed moment stall, but the minimum value of C_mM^2 reached at stall is in good agreement with experiment. On the tip section (Fig. 3(e)), the small negative C_mM^2 peak that occurs at $\psi = 290$ deg is well captured by the simulation. The slight differences between experiment and simulation in terms of minimum C_mM^2 and stall occurrence azimuth at the two last sections can be considered as acceptable since the experimental data are phase-averaged, whereas the numerical results are instantaneous values computed over one rotor revolution. As pointed out by several experimental works, phase-averaged measurements of stall phenomenon tend to reduce the intensity of the loads variations compared to instantaneous acquisitions (Refs. 33–35). Indeed, the flow during dynamic stall is never perfectly identical from one cycle to the other. Thus, the instantaneous loads slightly change between cycles, so that the phase-averaged data give lower oscillations than instantaneous ones. This might explain why the minimum pitching moment at stall is lower in the simulation than the experiment at the sections $r/R = 0.915$ and 0.975.

Thus, despite some slight differences, the main features of dynamic stall can be considered as fairly well captured by the simulation. However,

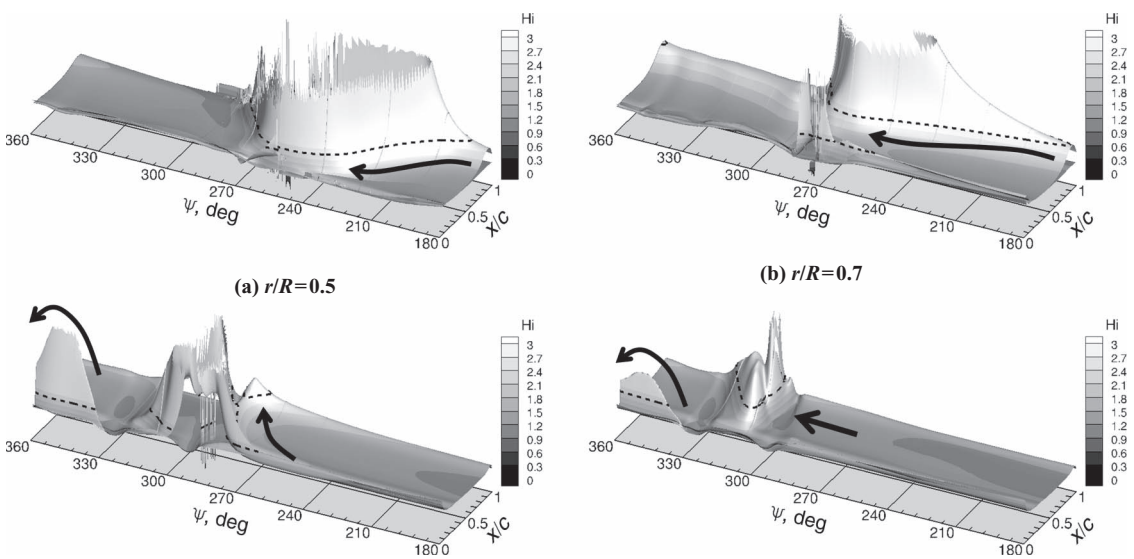
experimental measurements alone are not sufficient to describe the mechanisms that lead to stall. The objective of the next section is to use the rich amount of data provided by the simulation to augment the experimental measurements and identify the mechanisms responsible for stall.

Analysis

Identification of different stall regions

It is proposed here to extract, from the CFD results, the separation points for each blade radial station and at each time step, and finally draw a rotor map of the separation point to exhibit the existence, the localization, and the dynamics of the flow separation event(s) that occurred on the rotor blades during the simulation. To do that, a separation criterion is needed. Most of the literature dedicated to this problem has agreed that a suitable criterion to define attached and separated boundary layers should be related to the shape factor H_i, which is defined as the ratio of the momentum over displacement boundary layer thicknesses. In 2004, Castillo et al. proposed an expression of the critical shape factor at the separation point as $H_{\text{sep}} = 2.76 \pm 0.23$ (Ref. 36). This criterion provides satisfactory agreement with the experimental data of Refs. 37 and 38. Based on the CFD results of the present helicopter rotor simulation, the chordwise evolution of the shape factor with respect to the azimuthal position has been computed and is represented in Fig. 4 at four radial stations on the retreating side of the rotor disk. To ease the reading of the figures, the shape factor is only shown in a range between 0 and 3. The attached boundary layer is characterized by a low level of H_i, whereas the boundary layer is expected to be separated for the highest values. Figure 4 thus provides a qualitative view of the motion of the separation point along the chord as a function of the azimuthal position. Based on the conclusion of Ref. 36, the critical value $H_i = 2.7$ is shown in the dashed line as an indicator of the flow separation. In the innermost section ($r/R = 0.5$), the flow starts to separate at $\psi = 180$ deg from the trailing edge (TE) region (Fig. 4(a)). As ψ increases, the separation point moves progressively from the trailing to the LE as indicated by the black arrow in the figure. At $\psi = 250$ deg, the shape factor distribution indicates that the flow separation is expected to cover the entire blade chord. Then, from 250 to 290 deg of azimuth, the separation point moves progressively back to the TE. At $r/R = 0.7$ (Fig. 4(b)), the evolution of the shape factor is similar, except that the separation does not reach the LE but the midchord. In Fig. 4(c), the behavior is completely different. On this section ($r/R = 0.915$), high values of H_i suddenly appear at the LE for $\psi = 270$ deg and then rapidly spread to the TE as ψ increases. This motion of the separation point is illustrated by a black arrow on the right-hand side of the figure. A second event appears later, between $\psi = 330$ and 360 deg. It is pointed out by a second black arrow on the left-hand side of Fig. 4(c). In this event, the separation also starts from the LE region. But contrary to the previous separation event, the high values of H_i stays confined in a small region close to the LE without any spreading in the chordwise direction. This signifies that the flow reattaches to the wall before the TE, giving rise to a separation bubble. A subsequent analysis of the skin friction lines will confirm the observations based on the shape factor. On the tip section ($r/R = 0.975$), Fig. 4(d) exhibits a first separated flow region between $\psi = 300$ and 320 deg. Every chordwise position undergoes a growth of the shape factor at the same azimuthal angle, which suggests that this results from the radial contamination of the first flow separation that occurred previously on section $r/R = 0.915$. Figure 4(d) also exhibits a second flow separation event at the end of the fourth quadrant of the rotor disk and which is confined to the LE region and similar to what is observed for section $r/R = 0.915$ in Fig. 4(c).

To have an overall view of the dynamics of the flow separation(s) and to confirm the observations made with Fig. 4, the separation point expressed in the blade element local frame and noted x_{sep} has been computed along the blades for each time step using a criterion based on the shape factor. Following the range of the critical shape factor proposed by Castillo et al. in Ref. 36, separation point location is here defined as the chordwise x coordinate where the shape factor gets higher than the critical value $H_{\text{sep}} = 2.7$. This provides the rotor map of Fig. 5. The separation is expressed in the local frame and made nondimensional with respect to the chord, so that $x_{\text{sep}}/c = 0$ when the flow separation occurs at the LE and $x_{\text{sep}}/c = 1$ when the flow separation occurs at the TE, which means that the boundary layer remains attached all along the chord. Three distinct regions have been pointed out in Fig. 5, each one



(a) $r/R=0.5$ **(b) $r/R=0.7$**

(c) $r/R=0.915$ **(d) $r/R=0.975$**

Fig. 4. Azimuthal evolution of the boundary layer shape factor H_i with respect to the local chord coordinate x/c and the blade azimuth ψ at four different radius positions.

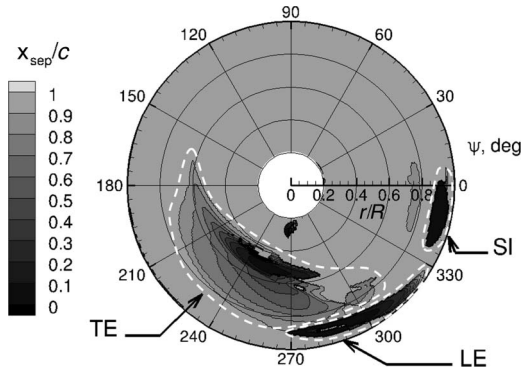


Fig. 5. Rotor map of the chordwise position of flow separation x_{sep}/c. Three different separation regions are bounded in white dashed lines. The TE stall region is referred to as TE, the LE stall region is referred to as LE, and the shock-induced separation region is referred to as SI.

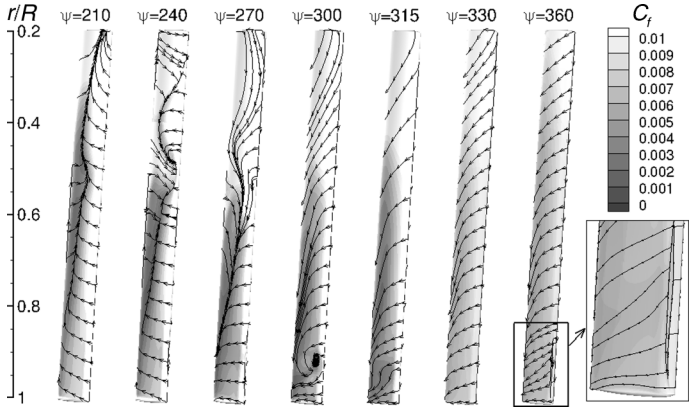


Fig. 6. Skin friction coefficient C_f and skin friction lines on the upper side of the retreating blade at different azimuthal positions ψ.

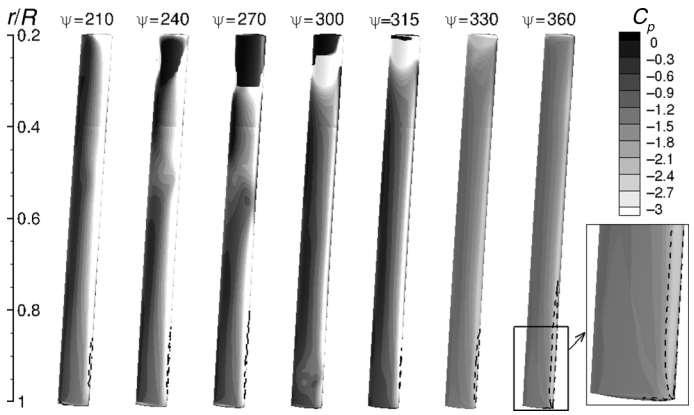


Fig. 7. Pressure coefficient C_p on the upper side of the retreating blade at different azimuthal positions ψ. The critical pressure coefficient where the local Mach number reaches one is represented by the dashed line.

being delimited by white dashed lines. The first region, called the TE stall region and referred to as TE, concerns the inboard part of the blade ($0.5 \leq r/R \leq 0.8$). It is mainly located in the third quadrant of the rotor disk. The second one, called the LE stall region and referred to as LE, is located in the outboard part of the blade ($0.9 \leq r/R \leq 0.95$) at the beginning of the fourth quadrant. Finally, a third region appears on the outboard part of the blade at the very end of the fourth quadrant. This last region is called the shock-induced separation region and referred to as SI thereafter.

In the TE stall region, the separation point first starts from the TE at the station $r/R = 0.6$ and $\psi = 170$ deg. Then, as ψ increases, the separation point moves upstream in the chordwise direction to finally reach the LE at station $r/R = 0.5$ and $\psi = 230$ deg. At the same time, the separation region also spreads in the spanwise direction, progressively contaminating the outboard part of the blade up to $r/R = 0.85$. This stall region presents three main features of the so-called TE stall as defined in the stall classification proposed by McCullough and Gault in Ref. 39. First, the flow separation moves continuously from the trailing to the LE. Second, the variations of lift and pitching moment at stall are smooth and continuous (Figs. 2(a) and 3(a)). Third, the airfoil located in this blade region ($r/R \leq 0.75$) has a high relative thickness of 13%. The skin friction lines shown in Fig. 6 confirm the conclusions deduced on the x_{sep} rotor map in Fig. 5. In the TE stall region, the skin friction lines draw a separation zone moving progressively from the TE to the LE as ψ increases. This is particularly visible for the snapshots $\psi = 210$ and 240 deg. The time variations of the shape factor previously observed in Figs. 4(a) and 4(b) exactly corresponds to this TE stall event.

The LE stall region in Fig. 5 concerns a thin radial portion of the blade around $r/R = 0.9$. Unlike the previous stall region, the flow abruptly separates here at the LE, which corresponds to the main feature of the so-called LE stall with respect to the categorization terminology of Ref. 39. The evolution of x_{sep}/c along the line $r/R = 0.9$ indeed undergoes a discontinuous change from 1 to 0 at $\psi = 270$ deg. Although, this LE stall region is nearby the TE stall region described previously, it clearly appears as an independent event and does not result from a contamination from this first stall region. This LE stall region also corresponds to the region where the variations of section lift and pitching moment are the strongest, both experimentally and numerically (Figs. 2(d) and 3(d)), which corresponds to the characteristics of a LE stall. The friction lines of Fig. 6 confirm that a localized separation cell at the leading of the blade for $0.85 \leq r \leq 0.95$ is observed at $\psi = 300$ deg. This separation cell leads to a decrease of the pressure in the LE region of the blade (Fig. 7) that causes a violent moment stall as observed at $r/R = 0.925$ in Fig. 3(d). To have an accurate description of the formation of this stall cell, a close-up view of the skin friction lines in the LE stall region between $\psi = 300$ deg and $\psi = 315$ deg is shown in Fig. 8. At $\psi = 270$ deg, a very thin separation region of less than 1% of chord appears at the LE. This structure is here too small to have an impact to the pressure coefficient. Then, the reattachment point progressively moves toward the TE as the azimuthal angle increases. At $\psi = 282$ deg, the separation zone extends to more than 30% of chord and the pitching moment stars to become negative (Fig. 3(d)). The stall cell goes on growing until $\psi = 294$ deg when $C_m M^2$ reaches its minimum. Then, the flow separation length starts to decrease. At $\psi = 312$ deg, it completely disappears, the pitching moment reaches zero and begins to get positive again as shown in Fig. 3(d). The last section ($r/R = 0.975$) also undergoes at the same azimuthal range some variations of lift and pitching moment due to stall but with a lower intensity (Fig. 3(e)). Friction lines of Fig. 8 show that this last radial section is contaminated at $\psi = 309$ deg by the stall cell that has appeared previously around the radial position $r/R = 0.9$. This contamination does not affect the LE at $r/R = 0.975$ but only the rear part of the blade, which explains why $C_m M^2$ variation at $r/R = 0.975$ is lower than $r/R = 0.915$ (Fig. 3(e)) and why the pitching moment is here almost never negative.

Finally, a third region, called the SI separation region, appears in Fig. 5. Here, the flow separates close to the LE of the blade tip

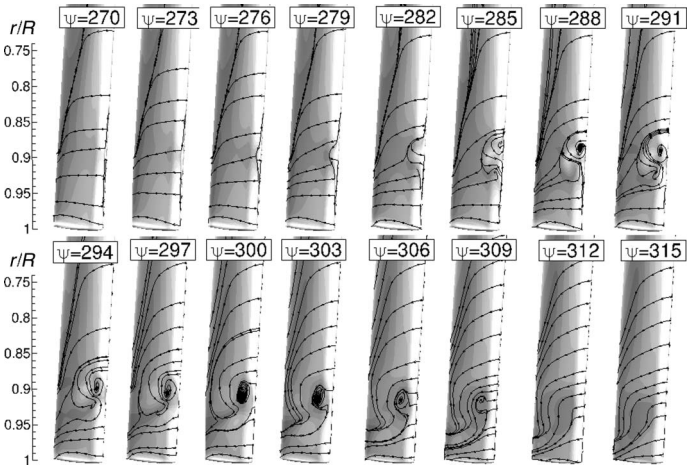


Fig. 8. Azimuthal evolution of skin friction lines in the FS2 region.

$(0.85 \leq r/R \leq 0.95)$ at an azimuthal station between $\psi = 340$ and 360 deg. In this region of the rotor disk, the pitching moment in Figs. 3(d) and 3(e), for both experiment and simulation, does not show any evidence of stall but only provides some small fluctuations that can be associated to this third separated flow region. A careful look at Fig. 6 confirms an abrupt change of friction lines direction very close to the LE that is more visible in the blade tip view on the bottom right of the figure. In this region of the rotor risk, the relative Mach number is high and close to the tip Mach number $M_{\text{tip}} = R\Omega/a = 0.646$. Furthermore, the aerodynamic angle of attack is expected to be high because of the simultaneous action of three effects. First, the torsion deformation angle reaches at this azimuth range its maximum positive value which is +1.5 deg (Fig. 9). Second, the positive lateral cyclic pitch angle, which reaches 2.26 deg (Table 3), is also expected to increase the angle of attack at this station of the rotor disk. Third, the blade is flapping down at this azimuthal range because of the Modane flapping law ($\beta_{1C} = -\theta_{1S}$), which also contributes to increase the angle of attack. Thus, the simultaneous effects of torsion, cyclic pitch, and blade flapping is suspected to increase the angle of attack of the blade tip sections at the end of the fourth quadrant of the rotor. This high angle of attack, combined with a relative Mach number higher than 0.6, gives rise to a strong shock wave close to the LE. The large supersonic region is indeed shown in dashed lines at $\psi = 360$ deg in Fig. 7. On the recompression edge of this supersonic region, a small separation bubble appears as indicated by the friction lines on the bottom right of Fig. 6. Because this separation is due to the shock wave, this region is called SI separation region and noted SI in Fig. 5. However, this region of the rotor disk cannot be, strictly speaking, named "stall region" unlike the other two. Although a small separation bubble clearly appears, it has no significant effect on the section normal forces and pitching moments that could be clearly related to a stall phenomenon. However, for higher rotor thrust coefficients or higher helicopter speed, this SI region could be strengthened and become a real third stall event, maybe similar to the stall regions observed by Bousman on the first quadrant of the rotor disk of the UH-60A in some flight conditions (Ref. 17). This should be confirmed by further numerical investigations.

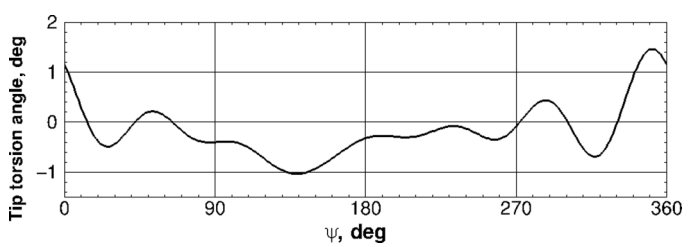


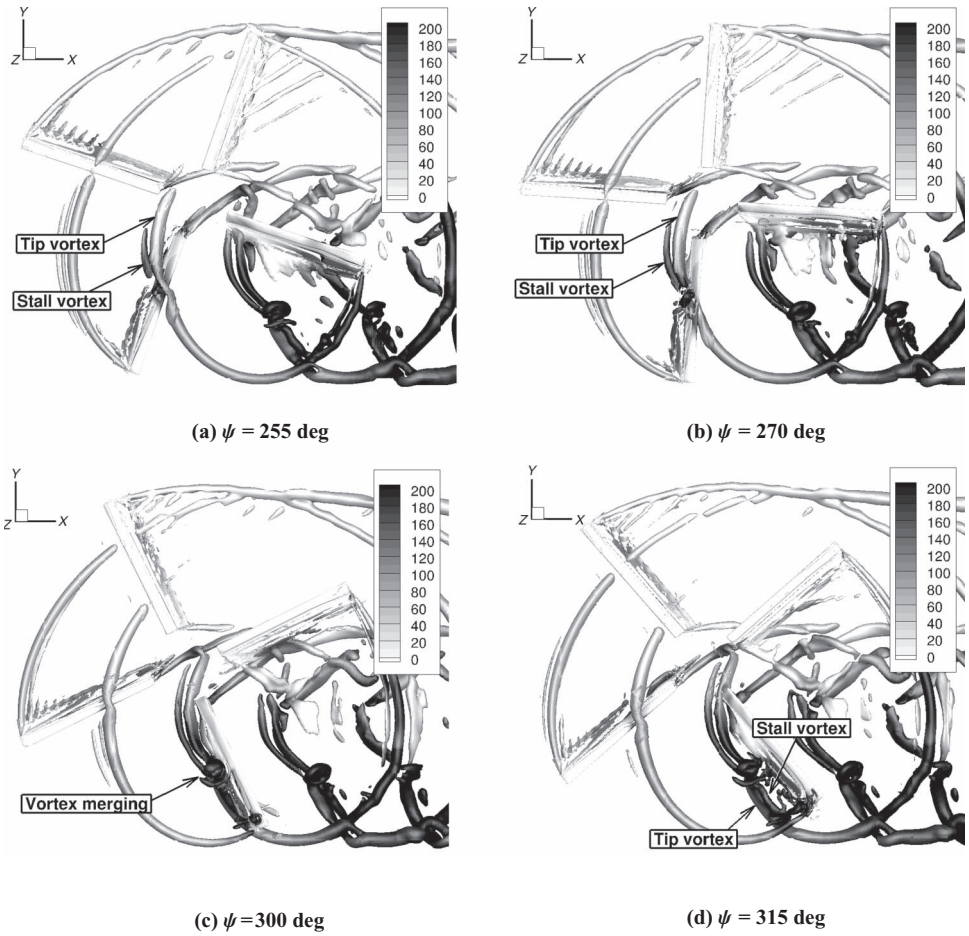
Fig. 9. Torsion deformation angle at the blade tip obtained with *elsA*/HOST coupling simulation.

Stall onset and blade–vortex interaction

The role of blade tip vortices in the dynamic stall process is investigated in this last part of the paper. Figure 10 shows an isocontour of Q-criterion for four azimuthal positions of the retreating blade: $\psi = 255$ deg, $\psi = 270$ deg, $\psi = 300$ deg, and $\psi = 315$ deg. The isocontour of Q-criterion reveals two different sources of vorticity inside the flow. The first one concerns the vortices that are generated at the blade tip by a primarily inviscid process. Thus, this kind of vortex is characterized by a low level of turbulent-to-laminar viscosity ratio μ_t/μ as shown in light gray in Fig. 10. A second kind of vorticity also emerges inside the flow. This one is due to the vortex shedding resulting from stall. In this case, the vortex contains high levels of μ_t/μ, since the vorticity is generated by the separation of the turbulent viscous boundary layer. These stall vortices appear in dark gray in Fig. 10.

At the azimuths $\psi = 255$ and 270 deg, TE stall occurs on the blade radial stations $0.4 \leq r/R \leq 0.8$. Figures 10(a) and 10(b) reveal that this stall event starts on the blade part that undergoes the impact of the tip vortex emitted by the previous blade. At these azimuthal stations, the blade–vortex interaction (BVI) is oblique. Flow separation occurs on the outboard part of the tip vortex impact where the local angle of attack is increased because of the vertical upwash velocity induced by the tip vortex. The stall vortex is thus emitted on the outboard part of the BVI region, which means that the stall vortex appears on the bottom left-hand side of the tip vortex in Figs. 10(a) and 10(b). The azimuthal angles $\psi = 300$ and 315 deg are related to the TE stall that occurs on the outboard part of the blade ($r/R \geq 0.9$). Figures 10(c) and 10(d) show that this stall event also coincides with the impact of the tip vortex coming from the previous blade. The difference with the previous stall event relies on the fact that the BVI is here almost parallel and not oblique. Thus, the stall vortex is emitted after the BVI, which corresponds to the right-hand side of the tip vortex in Figs. 10(c) and 10(d). Finally, the relative positions of the tip and stall vortices are opposite to each other between the TE stall region (inboard part of blade) and the LE stall region (outboard part of the blade). In between, the blade tip and stall vortices merge with each other in the region indicated in Fig. 10(c). This vortex-merging process distorts the classical helical shape of the blade tip vortex and gives rise to a vortex tangle, which is still visible in the rotor wake as shown in Fig. 10(d).

The TE and LE stall events that occur for this flight condition are thus undoubtedly triggered by a BVI phenomenon. This kind of scenario of dynamic stall in helicopter rotor environment is unusual in the literature. The classical description of dynamic stall, as it is generally assumed to occur on a helicopter rotor, is primarily due to the pitching and flapping motion, which makes the retreating blade reach such high angle of attack that dynamic stall occurs (Refs. 40–42). This idea led the helicopter community to investigate simplified configurations of oscillating pitching airfoils in uniform inflow which are indeed sufficient to reproduce the features of this kind of rotor stall scenario. BVI is also often neglected in the literature even when dynamic stall is addressed on rotating blades (Ref. 43), since this is largely accepted to have a minor effect. However, Gibertini and colleagues recently investigated experimentally the effect of BVI on an oscillating pitching airfoil, pointing out that BVI could trigger locally the stall onset on helicopter rotor blades (Refs. 44 and 45). The numerical investigation performed here proves that the impact of the tip vortex emitted by the previous blade indeed causes the beginning of



(a) ψ = 255 deg

(b) ψ = 270 deg

(c) ψ = 300 deg

(d) ψ = 315 deg

Fig. 10. Isocontour of Q-criterion ($Q = 5000\ \mathrm{s}^{-1}$) shaded by the turbulent-to-laminar viscosity ratio μ_t/μ.

stall. The bounded part of blade affected by this BVI thus undergoes a very rapid local increase of angle of attack which is added to its value, already high because of the pitching and flapping motion of the blade. This part of blade reaches higher angle of attack than its neighbors do. Thus, the stall onset is observed exactly where the tip vortex impacts the blade. This occurs first in the inboard part of the blade where the interaction is oblique. As the blade moves in azimuth, the position of the BVI moves from the inboard to the outboard parts of the blade. The stall onset along the blade radius then follows the motion of the BVI region, from the inboard to the outboard blade sections. This exactly corresponds to the radial motion of the separation point illustrated in Fig. 5 in the TE stall region. At the end of the BVI process, the tip vortex impact is no longer oblique but parallel to the blade, now causing the second stall referred to as the LE stall in Fig. 5 since the flow here separates abruptly from the LE.

The role of BVI in the stall onset is thus clearly highlighted for this particular rotor configuration. However, it does not prove that dynamic stall always occurs this way for every kind of helicopter rotor flight conditions. The stations on the rotor disk where BVI can occur and the intensity of this interaction depend on many parameters of the rotor flow: thrust coefficient, rotor shaft angle, advance ratio, etc. The wind tunnel test flow parameters presented in Table 1 are meant to be representative of a helicopter rotor in high-thrust forward flight at moderate advance ratio. However, if, for instance, a higher propulsive coefficient had been chosen, the rotor shaft angle would have been higher and the tip vortices would have been convected further downstream, away from the retreating blade, so that the BVI intensity would have been significantly reduced or even removed. Thus, it is not obvious whether, in this condition, stall would have occurred, and whether the impact of the tip vortices would have triggered the stall onset. Further numerical investigations will thus be necessary to clarify the role of BVI in the rotor stall process as a function of the rotor flow conditions.

Conclusions

The dynamic stall of the isolated 7A rotor in moderate-speed and high-thrust forward flight has been numerically investigated. A weak coupling strategy between ONERA CFD code *elsA* and Airbus Helicopters CA code HOST has been employed. Attention has been paid to the effect of time and space resolution on the quality of the numerical results, by basing this analysis on the requirements of previous numerical investigations of stall phenomena for simpler configurations of two-dimensional airfoils and three-dimensional finite span wings. The numerical simulation has provided very satisfactory results by comparison with the experimental data in terms of section normal forces and pitching moments. A thorough analysis of the flow separation has shown the existence of three different separated flow regions on the rotor disk. The first one concerns the inboard part of the blade, occurs in the third quadrant of the rotor disk, and is of TE stall type. The second one occurs close to the blade tip at the beginning of the fourth quadrant and is of LE stall type. In the third region, located on the blade tip at the end of the fourth quadrant, a small separation bubble appears at the root of a shock wave, but cannot be

related to a stall event since section normal forces and pitching moments do not exhibit strong variations associated with stall at these stations of the rotor disk. At last, the onset of stall has been clearly connected to the impact of the tip vortex generated by the previous blade. In the TE stall region the BVI is oblique, and in the LE stall region, where the effects of stall are the strongest, the BVI is parallel. This analysis points out that dynamic stall in helicopter environment can be triggered by a BVI. This dynamic stall scenario is unusual in the literature. Thus, further investigations will be necessary to determine and explain in which rotor conditions BVI can be a crucial element in the dynamic stall onset.

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